Database Lab : NoSQL Workload Classification & Storage Catalog Setup

Duration : 30 Min.

Scenario

As a Cloud Data Architect at NIIT, an enterprise microservice platform requires storage category mapping for 4 new workloads. You must establish a workload architecture catalog table in PostgreSQL and populate the recommended storage categories (Key-Value, Document, Wide-Column, Graph).

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **PostgreSQL Service:** `localhost:5432` | DB: `postgres` | User: `postgres` | Password: `PostgresPassword123!` | CLI: `psql`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Resource naming: - **Architecture Catalog Table:** `<prefix>_nosql_catalog`

Task Objectives

Perform the following configuration steps directly in PostgreSQL via `psql`:

1. Create Architecture Catalog Table

- Connect to `localhost:5432` with `psql`.
- Create table `<prefix>_nosql_catalog` with columns:
- `workload_name VARCHAR(100) PRIMARY KEY`
- `recommended_category VARCHAR(50) NOT NULL`
- `representative_engine VARCHAR(50) NOT NULL`
- `latency_sla VARCHAR(30) NOT NULL`

2. Populate Workload Architecture Classifications

- Insert the four enterprise workload mappings:
- `('User Session Token Cache', 'Key-Value', 'Redis', '< 1ms')`
- `('Product Catalog Dynamic Attributes', 'Document', 'MongoDB', '< 10ms')`
- `('Fleet IoT Sensor Time-Series', 'Wide-Column', 'Cassandra', '< 50ms')`
- `('Fraud Detection Multi-Hop Traversal', 'Graph', 'Neo4j', '< 25ms')`

3. Verify Catalog Consistency

- Run `SELECT * FROM <prefix>_nosql_catalog ORDER BY workload_name;` to confirm records are present.

Verification

Once completed, run the verification script:

```
python run.py
```

The verification system inspects the live PostgreSQL database table on `localhost:5432`.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	PostgreSQL Table (_nosql_catalog) Creation	4 Marks
TC2	NoSQL Category Mapping Population (4 Workloads)	3 Marks
TC3	Representative Engine & Latency SLA Accuracy	3 Marks

Total Score: 10 Marks

Important Notes

- Replace `<prefix>` with your candidate prefix.
- Ensure all 4 workload records are inserted.